Contact

ekashmirathore@gmail.com

www.linkedin.com/in/ekashmirathore (LinkedIn)

Top Skills

Water Treatment

Water Quality

Decarbonization

Languages

English

Hindi

Certifications

Stanford Ignite – Full-Time

Honors-Awards

Best Master Thesis of the Year 2017 at JNCASR

Winner of Falling Walls Lab India 2017

AIR 373, IIT-JAM

Gold Medalist in National Mathematics Olympiad Contest

29th Rank Holder in University of Delhi

Publications

Enhanced covalency and nanostructured-phonon scattering lead to high thermoelectric performance in n-type PbS

Reversible and efficient sequestration of Cs from water by layered metal thiophosphate, K0.48Mn0.76PS3.H2O

Enhanced Thermoelectric Performance in Mg2Si by Functionalized Co-Doping

Nature-Inspired Coral-like Layered [Co0.79Al0.21(OH)2(CO3)0.11]·mH2O for Fast Selective ppb Level Capture of Cr(VI) from Contaminated Water

Ekashmi Rathore, Ph.D.

I build personal apps with AI

San Francisco Bay Area

Summary

I am a scientist and innovation strategist passionate about translating deep technical insights into real-world impact.My work sits at the intersection of sustainability, resource recovery, and commercialization of emerging technologies. During my doctoral and postdoctoral research, I developed advanced materials for energy and environment application while collaborating with startups to move ideas from the lab toward real-world deployment.Most recently at Stanford, I supported commercialization and market-facing efforts for an early-stage technology focused on resource recovery from wastewater and green ammonia production. I conducted customer discovery with industrial stakeholders, evaluated deployment constraints, and helped shape strategy for bringing the technology to market.Earlier in my career, I worked closely with founders and multidisciplinary teams in startup environments translating complex science into practical solutions and strategic decisions.I’m particularly interested in solving large-scale challenges in:• Circular economy and resource recovery• Climate and sustainability technologies• Water, energy, agriculture and industrial decarbonization• Technology commercialization and go-to-market strategyAlongside my technical work, I’m deeply interested in community building, mental health, and leadership development, and I actively explore ways to bring people together around shared purpose.Currently, I am exploring opportunities in consulting, innovation strategy, and sustainability-focused problem solving, where I can apply analytical thinking and cross-disciplinary collaboration to tackle complex global challenges.If you’re working on impactful technologies or strategic challenges in sustainability, I’d love to connect.

Experience

Stanford University Graduate School of Business

Ignite Fellow

June 2026 - July 2026 (2 months)

Origin of ultra-low thermal conductivity in n-type cubic bulk AgBiS2: Soft Ag vibrations and local structural distortion induced by Bi 6s2 lone pair

Stanford University
Postdoctoral Scientist
April 2025 - April 2026 (1 year 1 month)
Stanford, California, United States

Electrochemical seperation

Magrathea
Research Engineer
October 2022 - July 2023 (10 months)
Oakland, California, United States

Electrolytic Technology Development: Contributed significantly to the advancement of innovative electrolytic technology, enabling carbon-neutral magnesium metal production from seawater and brines, aligning with the global transition toward electrification and decarbonization.
Process Design and Experimental Expertise: Led brine pilot processing project design, created BFD, PFD, P&IDS to ensure procurement readiness by utilizing draw.io software. Designed and conducted bench scale and pilot scale experiments based on M&E balance to optimize operation of brine processing pilot, demonstrated a hands-on approach to research and development.
In-House Analytical Technique Development: Spearheaded the development of in-house wetlab space, procure & setup analytical instruments, develop analytical methods, design SOPs, to support pilot plant operations and improve process
efficiency.
Problem-Solving and Continuous Improvement: Applied root cause and failure analysis methodologies to troubleshoot and address challenges in the research and development process, contributing to continuous improvement efforts.

JNCASR
Research And Development Scientist
August 2017 - December 2021 (4 years 5 months)
Bengaluru Area, India

Experience gained as PhD (Area:- Exploring 3D & 2D- layered Chalcogenides for thermoelectric, waste water treatment & solar cells)

• Leading a team of 3 members on chemical research involving searching & developing new materials for water purification & thermoelectrics as a solid-state/ material chemist.

• Demonstrated problem-solving, leadership, and project management skills by successfully completing execution of 10+ projects as a technical expert, publishing in 7 journals, and contributing in bringing #50 Lacs grant funding in Lab by DST, India.
• Executed systemization, strategic planning resulting in the development of 5 new products for water purification and thermoelectrics applications by complying standard operating procedures (SOP).
• Disseminated research results in 20+ international and national conference as a proficient communicator (verbal, virtual and written).
• Accumulated interpersonal skills and cross-functional research data analysis skill by collaborating with scientists from 4 different research groups (IISc Bangalore & IIT Mandi, India; Justus-Liebig-Universität Giessen, Germany; J-PARC, Ibaraki, Japan).
• Gained knowledge in industry trends and commercial acumen as an in-charge of ordering lab resources for 2 years (chemicals, glassware, and other equipment, 2-3 times in a month).
• Acquired business development through interacting with vendors and institution financial specialist while maintaining lab budget (reduced project cost by ¼th of the original amount).
• Nurtured safety protocols, managed glove-box, gas cylinders, and trained annually by EHS organization.

RainIons
Research Consultant
June 2021 - August 2021 (3 months)
California, United States

-Validating Start-up idea for capturing carbon dioxide from the exhaust of vehicles and cooling towers and transformation into fuel with data analysis and research, directly managed by CTO and CEO.
-Helping to find mechanism for carbon dioxide capture and transformation into fuel.
-Science communication- making visual science art and writing articles related to CO2 capture

Khetee
Curriculum Designer & Digitization Expert
August 2020 - October 2020 (3 months)
Bihar, India

Volunteering as Educational Intern at Khetee, NGO based in Bihar, to help the farmers practice Regenerative Agroforestry (Goal- Figuring OUT)

• Designed curriculum for practicing regenerative agroforestry by interacting with 10+ farmers, learned about crops, management, processing in agriculture fields
• Digitized NGO's social media presence at Facebook, Instagram, LinkedIn, increased followers organically, created awareness, connected like-minded users interested in farming
• Managed Instagram handles, designed content, and executed global Instagram strategy, and utilized the data to understand what worked to increase the followers by 40 % through regular engagement with the user by organizing contests.

Jagriti Yatra
Change Maker- Yatri
December 2018 - January 2019 (2 months)
Mumbai, Maharashtra, India

Mini-MBA: 15 days, 8000 km national odyssey to meet the role models - social and business entrepreneurs - of the country to build India through enterprise. Trained in entrepreneurship, worked on a team-building project to provide better user experience in villages, figured out problem statements, pitched a business idea to impact 2000+ lives.

Runner up in Jagriti Yatra Business Gyan Tree-Open Innovation Challenge for building social entrepreneurship in Water, Sanitation and Hygiene (WASH) through BASF-JAGRITI initiative.

Falling Walls Foundation
Innovator | Science Communicator
November 2017 - November 2017 (1 month)
Berlin Metropolitan Area

• Represented India in Falling Walls Lab, Berlin Germany, and pitched water purification research innovation at Falling Walls Lab 2017, Berlin, Germany, among 100 innovators from different countries, in just 3 minutes!
" Breaking the wall of slow death: Safer Water, Smarter Tomorrow"

Education

Stanford University Graduate School of Business

Stanford Ignite Program, Entrepreneurship/Entrepreneurial Studies · (May 2026 - June 2026)

Jawaharlal Nehru Centre for Advanced Scientific Research, Bangalore

MS and Ph.D., Materials Science · (2014 - 2021)

Hindu College, University of Delhi

B.Sc. (Hons.) , Chemistry · (2011 - 2014)